

QMWS Spring 2026: Introduction to Python for Data Analysis

Day One: Foundations

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Section 1

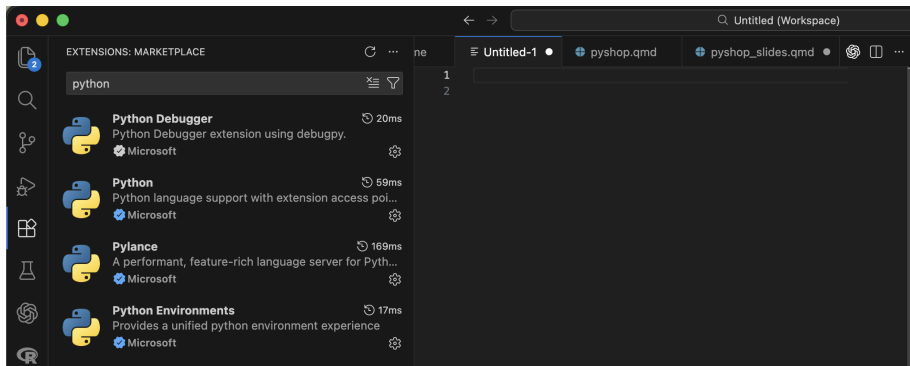
Installation

Visual Studio (VS) Code

- Installation link: <https://code.visualstudio.com/download>
- Integrated Development Environment (IDE)
- Can use it to code in many different languages
- Commonly used in both academia and industry

Visual Studio (VS) Code

- Once installed, go to the “Extensions” tab and search “Python”
- Install the below extensions:



- A tool for configuring Python **environments**
 - More often than not, you need to install **packages** that allow you to code more easily/efficiently
 - Allows for version control
 - Issues can arise if environments are not used, for example:
 - If you install two packages that conflict with one another
 - If you install a package that requires a specific version of Python
 - Anaconda allows these issues to be isolated to a specific environment, rather than affecting your whole Python installation
 - Especially useful if you have more than one project going on

- Installation link: <https://www.anaconda.com/download>
- The simplest way to use Anaconda is via the **Navigator**
- To install:
 - 1 Make an account
 - 2 Install the full distribution via the “Graphical Installer”

- Installation link: <https://www.anaconda.com/download>
- On Mac:



Choose Your Download

Windows

Mac

Linux

Anaconda Distribution

Complete package with 8,000+ libraries, Jupyter, JupyterLab, and Spyder IDE. Everything you need for data science.

↓ [64-Bit \(Apple silicon\) Graphical Installer](#)

↓ [64-Bit \(Apple silicon\) Command Line Installer](#)

Miniconda

Minimal installer with just Python, Conda, and essential dependencies. Install only what you need.

↓ [64-Bit \(Apple silicon\) Graphical Installer](#)

↓ [64-Bit \(Apple silicon\) Command Line Installer](#)

↓ [64-Bit \(Intel chip\) Graphical Installer](#)

↓ [64-Bit \(Intel chip\) Command](#)

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Section 2

Setting Up a Conda Environment

Section 3

The Basics

- In programming, a variable is a space in computer memory where values may be stored

Defining and Printing Variables

```
x = 67
```

```
print(x)
```

```
67
```

Defining and Printing Variables

```
a = 1
```

```
b = 2
```

```
print(a + b)      # * for multiplication, / for division
```

```
3
```

Note

- Use # to make a **comment**

Variable Types

- ❶ `int` (integer; “whole number”)
 - less memory
 - 1 is an `int`
- ❷ `float` (real; “decimal number”)
 - 1.2 is a `float`
 - 1.0 is a `float`
- ❸ `str` (string; characters/words with no numerical meaning)
 - “Hello world!” is a string
 - “416-676-6767” is also a string
- ❹ `bool` (boolean; can be `True` or `False`; more on this later)
- ❺ `None`
 - A special type of variable that essentially means “nothing meaningful is stored”
 - Useful as a placeholder

Checking and Converting Between Variable Types

```
x = 67
```